

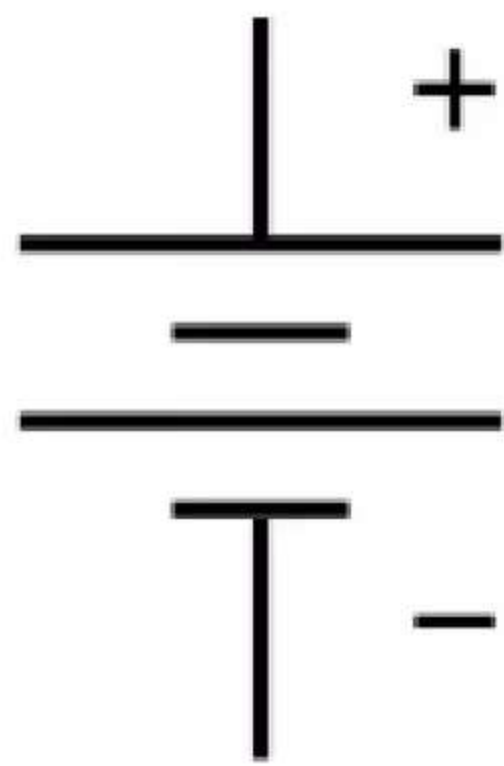
Battery



Various cells and batteries (top-left to bottom-right): two AA, one D, one handheld ham radio battery, two 9-volt, two AAA, one C, one camcorder battery, one cordless phone battery.

An electrical **battery** is one or more **electrochemical cells** that convert stored chemical **energy** into electrical energy.^[1] Since the invention of the first battery (or "**voltaic pile**") in 1800 by **Alessandro Volta**, batteries have become a common power source for many household and industrial applications

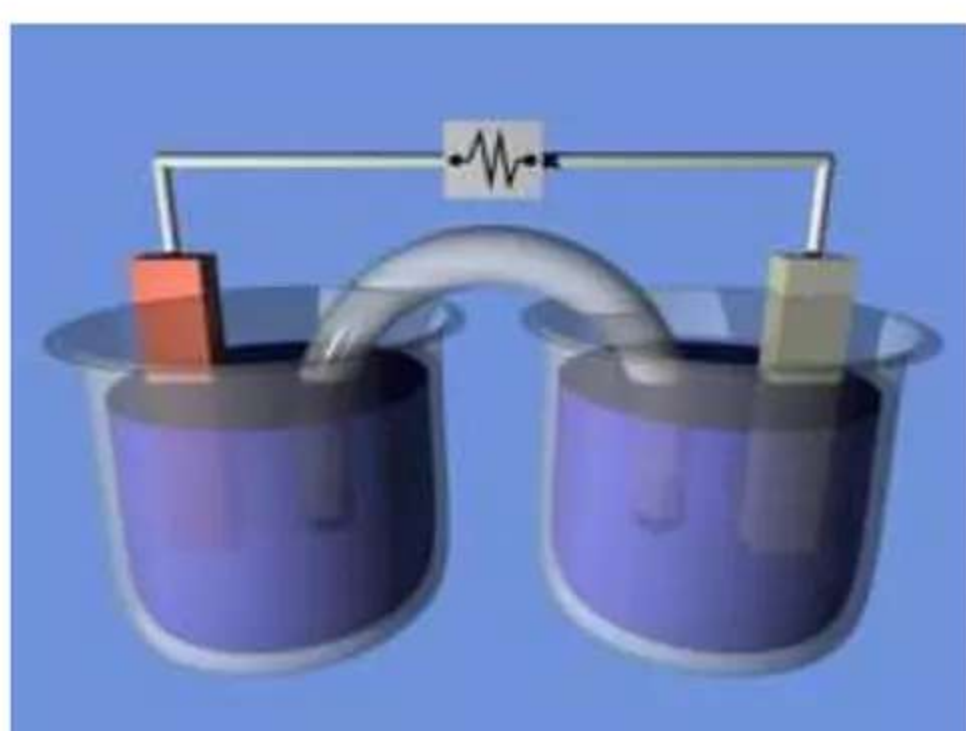
There are two types of batteries: **primary batteries** (disposable batteries), which are designed to be used once and discarded, and **secondary batteries** (rechargeable batteries), which are designed to be recharged and used multiple times. Batteries come in many sizes, from miniature cells used to power **hearing aids** and wristwatches to battery banks the size of rooms that provide standby power for **telephone exchanges** and computer **data centres**.



The **symbol** for a battery

Strictly, a battery is a collection of multiple electrochemical cells, but in popular usage *battery* often refers to a single cell.^[1] For example, a 1.5-volt AAA battery is a single 1.5-volt cell, and a 9-volt battery has six 1.5-volt cells in series.

Principle of operation



A voltaic cell for demonstration purpose. In this example the two half-cells are linked by a **salt bridge** separator that permits the transfer of ions, but not water molecules.

A battery is a device that converts chemical energy directly to electrical energy.^[22] It consists of a number of voltaic cells; each voltaic cell consists of two **half cells** connected in series by a conductive electrolyte containing anions and cations. One half-cell includes electrolyte and the electrode to which **anions** (negatively charged ions) migrate, i.e., the **anode** or negative electrode; the other half-cell includes electrolyte and the electrode to which **cations** (positively charged ions) migrate, i.e., the **cathode** or positive electrode. In the **redox** reaction that powers the battery, cations are reduced (electrons are added) at the cathode, while anions are oxidized (electrons are removed) at the anode.^[23] The electrodes do not touch each other but are electrically connected by the **electrolyte**. Some cells use two half-cells with different electrolytes. A separator between half cells allows ions to flow, but prevents mixing of the electrolytes.

The voltage developed across a cell's terminals depends on the energy release of the chemical reactions of its electrodes and electrolyte. Alkaline and **carbon-zinc** cells have different chemistries but approximately the same emf of 1.5 volts; likewise **NiCd** and **NiMH** cells have different chemistries, but approximately the same emf of 1.2 volts.^[29] On the other hand the high electrochemical potential changes in the reactions of **lithium** compounds give lithium cells emfs of 3 volts or more.^[30]

Categories and types of batteries



From Top to bottom: a large 4.5-volt (3R12) battery, a *D Cell*, a *C cell*, an *AA cell*, an *AAA cell*, an *AAAA cell*, an *A23 battery*, a 9-volt *PP3 battery*, and a pair of *button cells* (CR2032 and LR44).

Batteries are classified into two broad categories, each type with advantages and disadvantages.^[31]

- *Primary* batteries irreversibly (within limits of practicality) transform chemical energy to electrical energy. When the initial supply of reactants is exhausted, energy cannot be readily restored to the battery by electrical means.^[32]
- *Secondary* batteries can be recharged; that is, they can have their chemical reactions reversed by supplying electrical energy to the cell, restoring their original composition.^[33]

Primary batteries

Primary cell

Primary batteries can produce current immediately on assembly. Disposable batteries are intended to be used once and discarded. These are most commonly used in portable devices that have low current drain, are only used intermittently, or are used well away from an alternative power source, such as in alarm and communication circuits where other electric power is only intermittently available. Disposable primary cells cannot be reliably recharged, since the chemical reactions are not easily reversible and active materials may not return to their original forms. Battery manufacturers recommend against attempting to recharge primary cells.^[35]

Common types of disposable batteries include [zinc-carbon batteries](#) and [alkaline batteries](#). In general, these have higher [energy densities](#) than rechargeable batteries,^[36] but disposable batteries do not fare well under high-drain applications with [loads](#) under 75 [ohms](#) (75 Ω).^[31]

Secondary batteries

Rechargeable battery

Secondary batteries must be charged before use; they are usually assembled with active materials in the discharged state. Rechargeable batteries or [secondary cells](#) can be recharged by applying electric current, which reverses the [chemical reactions](#) that occur during its use. Devices to supply the appropriate current are called chargers or rechargers.

The oldest form of rechargeable battery is the [lead-acid battery](#).^[37] This battery is notable in that it contains a liquid in an unsealed container, requiring that the battery be kept upright and the area be well ventilated to ensure safe dispersal of the [hydrogen](#) gas produced by these batteries during overcharging. The lead-acid battery is also very heavy for the amount of electrical energy it can supply. Despite this, its low manufacturing cost and its high surge current levels make its use common where a large capacity (over approximately 10 Ah) is required or where the weight and ease of handling are not concerns.

A common form of the lead-acid battery is the modern [car battery](#), which can, in general, deliver a peak current of 450 [amperes](#).^[38] An improved type of liquid electrolyte battery is the sealed valve regulated lead acid ([VRLA](#)) battery, popular in the automotive industry as a replacement for the lead-acid wet cell. The VRLA battery uses an immobilized sulfuric acid

electrolyte, reducing the chance of leakage and extending shelf life.^[39] VRLA batteries have the electrolyte immobilized, usually by one of two means:

- *Gel batteries* (or "gel cell") contain a semi-solid electrolyte to prevent spillage.
- *Absorbed Glass Mat* (AGM) batteries absorb the electrolyte in a special fiberglass matting.

Other portable rechargeable batteries include several "dry cell" types, which are sealed units and are, therefore, useful in appliances such as [mobile phones](#) and [laptop computers](#). Cells of this type (in order of increasing [power density](#) and cost) include [nickel-cadmium](#) (NiCd), [nickel-zinc](#) (NiZn), [nickel metal hydride](#) (NiMH), and [lithium-ion](#) (Li-ion) cells.^[40] By far, Li-ion has the highest share of the dry cell rechargeable market.^[3] Meanwhile, NiMH has replaced NiCd in most applications due to its higher capacity, but NiCd remains in use in [power tools](#), [two-way radios](#), and [medical equipment](#).^[3] NiZn is a new technology that is not yet well established commercially.

Recent developments include batteries with embedded functionality such as [USBCELL](#), with a built-in charger and [USB](#) connector within the AA format, enabling the battery to be charged by plugging into a USB port without a charger,^[41] and [low self-discharge](#) (LSD) mix chemistries such as Hybrio,^[42] ReCyko,^[43] and Eneloop,^[44] where cells are precharged prior to shipping.

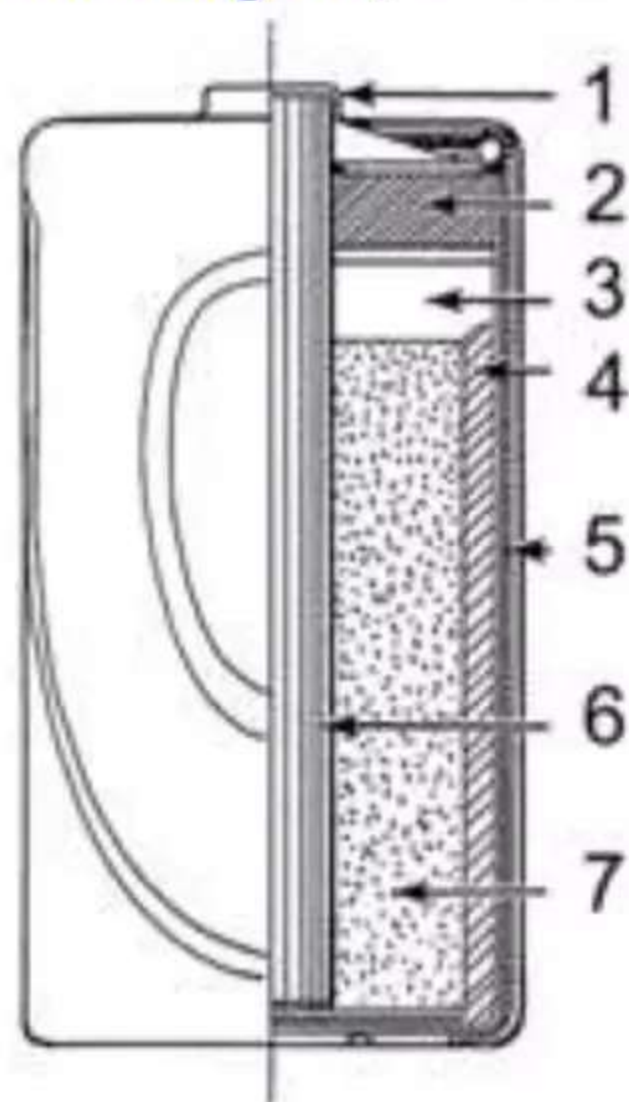
Battery cell types

There are many general types of electrochemical cells, according to chemical processes applied and design chosen. The variation includes [galvanic cells](#), [electrolytic cells](#), [fuel cells](#), [flow cells](#) and voltaic piles.^[45]

Wet cell

A *wet cell* battery has a liquid [electrolyte](#). Other names are *flooded cell*, since the liquid covers all internal parts, or *vented cell*, since gases produced during operation can escape to the air. Wet cells may be [primary cells](#) (non-rechargeable) or [secondary cells](#) (rechargeable). Wet cells are still used in [automobile batteries](#) and in industry for standby power for [switchgear](#), telecommunication or large [uninterruptible power supplies](#), but in many places

batteries with [gel cells](#) have been used instead. These applications commonly use lead-acid or [nickel-cadmium](#) cells.



Dry cell

Line art drawing of a dry cell:

1. brass cap, 2. plastic seal, 3. expansion space, 4. porous cardboard, 5. zinc can, 6. carbon rod, 7. chemical mixture.

A *dry cell* has the electrolyte immobilized as a paste, with only enough moisture in the paste to allow current to flow. As opposed to a wet cell, the battery can be operated in any random position, and will not spill its electrolyte if inverted.

A common dry cell battery is the [zinc-carbon battery](#), using a cell sometimes called the dry [Leclanché cell](#), with a nominal voltage of 1.5 [volts](#), the same nominal voltage as the [alkaline battery](#) (since both use the same [zinc-manganese dioxide](#) combination).

The makeup of a standard dry cell is a [zinc](#) anode (negative pole), usually in the form of a cylindrical pot, with a [carbon](#) cathode (positive pole) in the form of a central rod. The electrolyte is [ammonium chloride](#) in the form of a paste next to the zinc anode. The remaining space between the electrolyte and carbon cathode is taken up by a second paste consisting of ammonium chloride and manganese dioxide, the latter acting as a [depolariser](#).

Molten salt

A [molten salt battery](#) is a primary or secondary battery that uses a molten [salt](#) as its electrolyte. Their [energy density](#) and [power density](#) give them potential for use in [electric vehicles](#), but they must be carefully insulated to retain heat.

Fastest charging, largest, and lightest batteries

[Lithium iron phosphate \(LiFePO₄\) batteries](#) are the fastest charging and discharging, next to [supercapacitors](#).^[49] The world's largest battery is in [Fairbanks, Alaska](#), composed of [Ni-Cd cells](#).^[50] [Sodium-sulfur batteries](#) are being used to store [wind power](#).^[51] [Lithium-sulfur batteries](#) have been used on the longest and highest solar powered flight.^[52] The speed of recharging for lithium-ion batteries may be increased by manipulation.^[53]

Battery cell performance

A battery's characteristics may vary over load cycle, over charge cycle, and over lifetime due to many factors including internal chemistry, [current](#) drain, and temperature.

Improving Battery Life

The simplest and most obvious way of getting the maximum life out of a battery is to ensure that it always works well within its designed operating limits. There are however some further actions which can be taken to increase the battery life.

1.0 Charging

Most of the battery failures are due to inappropriate charging. The use of intelligent chargers and safety systems which prevent the connection of unapproved chargers to the battery may not extend battery life but at least they can prevent it from being cut short.

Battery Shelf Life: usually considered as having only 80% of its initial capacity

Battery Cycle Life is defined as the number of complete charge - discharge cycles a battery can perform before its nominal capacity falls below 80% of its initial rated capacity. Lifetimes of 500 to 1200 cycles are typical. The actual ageing process results in a gradual reduction in capacity over time. When a cell reaches its specified lifetime it does not stop working suddenly. The ageing process continues at the same rate as before so that a cell whose capacity had fallen to 80% after 1000 cycles will

probably continue working to perhaps 2000 cycles when its effective capacity will have fallen to 60% of its original capacity.

An alternative measure of cycle life is based on the internal resistance of the cell. In this case the cycle life is defined as the number of cycles the battery can perform before its internal resistance increases by an agreed amount., usually 1.3 times or double its initial value when new.

2.0 Battery Management

Battery management is essentially the method of keeping the cells within their desired operating limits during both charging and discharging either by controlling the load on the battery or by isolating the battery from the load if the load cannot be controlled.

3.0 Thermal Management

Extremes of temperature are battery killers. Both heating and cooling may be needed to keep the battery within its recommended operating limits. See

4.0 Load Sharing

For pulsed applications the peak load on the battery can be reduced by placing a large value capacitor in parallel with the battery. Energy for large instantaneous loads is supplied by the capacitor effectively reducing the duty cycle and stress on the battery. The capacitor recharges during the quiescent periods. Claims of a sixty percent increase in cycle life are made for this technique. .

5.0 Demand Management

The "effective" life of a battery in a particular application can also be extended by controlling the load which the application places on the battery. This does not actually improve the battery performance; instead it reduces the load that the battery has to supply.

6.0 Premature Death

The most likely cause of premature failure of a battery is abuse, subjecting a battery to conditions for which it was never designed.

Apart from obvious physical abuse, the following examples should also be considered abuse, whether deliberate, inadvertent or through poor maintenance disciplines.

- Drawing more current than the battery was designed for or short circuiting the battery.
- Using undersized batteries for the application.
- Circuit or system designs which subject the battery to repeated "coup de fouet" (whiplash) effects. This effect is a temporary, severe voltage drop which occurs when a heavy load is suddenly placed on the battery and is caused by the inability of the rate of the chemical action in the battery to accommodate the instantaneous demand for current.
- Operating or storing the battery in too high or too low ambient temperatures.
- Using chargers designed for charging batteries with different cell chemistry.
- Overcharging - either to too high a voltage or for too long a period.
- Over-discharging - allowing the battery to become completely discharged.
- In aqueous batteries - allowing electrolyte level to fall below the recommended minimum.
- In aqueous batteries - topping up with tap water instead of distilled water (or inappropriate electrolyte).
- Subjecting the battery to excessive vibration or shock.

Battery designers try to design out the possibility of abuse wherever possible but ultimately the life of the battery is in the user's hands.

Hazards

Explosion

A battery explosion is caused by the misuse or malfunction of a battery, such as attempting to recharge a primary (non-rechargeable) battery,^[70] or [short circuiting](#) a battery.^[71] With car batteries, explosions are most likely to occur when a short circuit generates very large currents

When a battery is recharged at an excessive rate, an explosive gas mixture of hydrogen and oxygen may be produced faster than it can escape from within the walls of the battery, leading to pressure build-up and the possibility of bursting of the battery case. In extreme cases, the battery acid may spray violently from the casing of the battery and cause injury. Overcharging—

That is, attempting to charge a battery beyond its electrical capacity—can also lead to a battery explosion, leakage, or irreversible damage to the battery. It may also cause damage to the charger or device in which the overcharged battery is later used. In addition, disposing of a battery in fire may cause an explosion as steam builds up within the sealed case of the battery.^[71]

Leakage



Leaked alkaline battery.

Many battery chemicals are corrosive, poisonous, or both. If leakage occurs, either spontaneously or through accident, the chemicals released may be dangerous.

For example, disposable batteries often use zinc "can" as both a reactant and as the container to hold the other reagents. If this kind of battery is run all the way down, or if it is recharged after running down too far, the reagents can emerge through the cardboard and plastic that forms the remainder of the container. The active chemical leakage can then damage the equipment that the batteries were inserted into. For this reason, many electronic device manufacturers recommend removing the batteries from devices that will not be used for extended periods of time.

Environmental concerns

The widespread use of batteries has created many [environmental concerns](#), such as toxic metal pollution.^[72] Battery manufacture consumes resources and often involves hazardous chemicals. Used batteries also contribute to [electronic waste](#).

Battery chemistry

Primary battery chemistries

Chemistry	Nominal Cell Voltage	Specific Energy [MJ/kg]	Elaboration
Zinc-carbon	1.5	0.13	Inexpensive.
Zinc-chloride	1.5		Also known as "heavy duty", inexpensive.
Alkaline (zinc-manganese dioxide)	1.5	0.4-0.59	Moderate energy density. Good for high and low drain uses.
Oxy nickel hydroxide (zinc-manganese dioxide/oxy nickel hydroxide)	1.7		Moderate energy density. Good for high drain uses
Lithium (lithium-copper oxide) Li-CuO	1.7		No longer manufactured. Replaced by silver oxide (IEC-type "SR") batteries.
Lithium (lithium-iron disulfide) LiFeS ₂	1.5		Expensive. Used in 'plus' or 'extra' batteries.
Lithium (lithium-manganese dioxide) LiMnO ₂	3.0	0.83-1.01	Expensive. Only used in high-drain devices or for long shelf life due to very low rate of self discharge. 'Lithium' alone usually refers to this type of chemistry.
Mercury oxide	1.35		High drain and constant voltage. Banned in most countries because of health concerns.
Zinc-air	1.35-1.65	1.59 ^[82]	Mostly used in hearing aids.
Silver-oxide (silver-zinc)	1.55	0.47	Very expensive. Only used commercially in 'button' cells.

Rechargeable battery chemistries

Chemistry	Cell Voltage	Specific Energy [MJ/kg]	Comments
NiCd	1.2	0.14	Inexpensive. High/low drain, moderate energy density. Can withstand very high discharge rates with virtually no loss of capacity. Moderate rate of self discharge. Reputed to suffer from memory effect (which is alleged to cause early failure). Environmental hazard due to Cadmium - use now virtually prohibited in Europe.
Lead acid	2.1	0.14	Moderately expensive. Moderate energy density. Moderate rate of self discharge. Higher discharge rates result in considerable loss of capacity. Does not suffer from memory effect . Environmental hazard due to Lead. Common use - Automobile batteries
NiMH	1.2	0.36	Inexpensive. Performs better than alkaline batteries in higher drain devices. Traditional chemistry has high energy density, but also a high rate of self-discharge. Newer chemistry has low self-discharge rate , but also a ~25% lower energy density. Very heavy. Used in some cars.
NiZn	1.6	0.36	Moderately inexpensive. High drain device suitable. Low self-discharge rate. Voltage closer to alkaline primary cells than other secondary cells. No toxic components. Newly introduced to the market (2009). Has not yet established a track record. Limited size availability.
Lithium ion	3.6	0.46	Very expensive. Very high energy density. Not usually available in "common" battery sizes (but see RCR-V3 for a counter-example). Very common in laptop computers, moderate to high-end digital cameras and camcorders, and cell phones. Very low rate of self discharge. Volatile: Chance of explosion if short circuited, allowed to overheat, or not manufactured with rigorous quality standards.

Reference: <http://www.mpoweruk.com/life.htm>